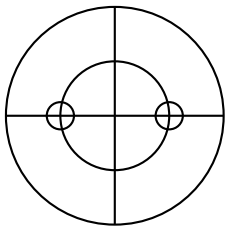
Dutton's Unified SEFI

0.5cm]**Single-Entity Field Interpretation**

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Date: August 2026

Abstract

Dutton's Unified SEFI proposes that all electrons and positrons are manifestations of a single underlying entity whose worldline spans all possible spacetime paths. The dense superposition of this worldline appears as a continuous field, reproducing the electron field of quantum field theory. Quantum interference arises from overlapping contributions of different segments of the entity's path structure, while measurement corresponds to selecting a subset of paths, producing a localized particle event. The model is formulated using relativistic worldline actions, Grassmann spin variables, electromagnetic gauge coupling, presence-density mapping, and the Dirac field in curved spacetime. SEFI preserves all predictions of quantum mechanics and quantum field theory while offering a unified ontological interpretation of particle identity, superposition, and measurement. Geometry is introduced as the natural language of the entity's worldline, unifying superposition, interference, identity, and entanglement as geometric consequences of one underlying object.

1 Introduction

Quantum mechanics and quantum field theory (QFT) successfully predict experimental outcomes, yet the ontological status of particles, fields, and superposition remains conceptually unresolved. Standard interpretations treat superposition as a mathematical feature of the wavefunction, while leaving open the question of what, if anything, is physically "in superposition."

Dutton's Unified SEFI (Single-Entity Field Interpretation) proposes a different starting point: all electrons and positrons are manifestations of a single underlying entity whose worldline spans all allowed spacetime trajectories. The electron field is the visible projection of this entity's dense worldline structure. Superposition, interference, and measurement are not separate phenomena; they are geometric consequences of the entity's worldline.

This framework is compatible with QFT, path integrals, and relativistic field theory, but adds a clear ontology: there is one entity, one worldline structure, and many local manifestations.

2 Core Postulates of SEFI

2.1 P1 — Single Underlying Entity

All electrons and positrons are local manifestations of a single fundamental entity. This entity is not a point particle but a distributed worldline structure embedded in spacetime.

2.2 P2 — Bidirectional Time Segments

Forward-time segments of the entity's worldline appear as electrons; backward-time segments appear as positrons. Time reversal is an orientation change along the worldline.

2.3 P3 — Dense Spacetime Weaving

The entity's worldline is not a single classical curve. It is a dense, possibly fractal-like ensemble of trajectories that collectively span all allowed paths in spacetime. This dense weaving produces the appearance of a continuous field.

2.4 P4 — Emergent Field Appearance

The observed electron field is the presence density generated by the entity's full worldline ensemble. The field is not fundamental; it is the projection of the entity's geometry.

2.5 P5 — Interference as Path Overlap

Quantum interference arises from overlapping contributions of different segments of the entity's worldline. Where paths overlap, amplitudes combine; where they cancel, destructive interference occurs.

2.6 P6 — Measurement as Path Selection

Measurement imposes boundary conditions that restrict the allowed worldline segments. Collapse is not destruction of the entity; it is the selection of one segment (or subset of segments) from the ensemble.

2.7 P7 — Ontological Unity of Particle Identity

Identical particle properties arise because all electrons and positrons are local manifestations of the same underlying entity. Identity is global and geometric, not local and discrete.

3 Mathematical Framework

3.1 Worldline Foundation

Let spacetime be a manifold M with metric $g_{\mu\nu}$. The entity's worldline structure is represented by an ensemble of paths $\gamma(\tau)$ embedded in M. A classical relativistic worldline action is:

$$S_{\text{WL}} = -m \int d\tau \sqrt{-g_{\mu\nu}(\gamma(\tau))\dot{\gamma}^{\mu}(\tau)\dot{\gamma}^{\nu}(\tau)}.$$

3.2 Spin Structure

Spin degrees of freedom are encoded using Grassmann variables $\Psi_a(\tau)$:

$$S_{\text{spin}} = \int d\tau \left[\frac{i}{2}\Psi_a(\tau)\dot{\Psi}^a(\tau)\right].$$

3.3 Electromagnetic Gauge Coupling

Coupling to an electromagnetic potential $A_\mu(x)$ is:

$$S_{\text{EM}} = q \int d\tau\, \dot{\gamma}^{\mu}(\tau) A_{\mu}(\gamma(\tau)).$$

3.4 Total Worldline Action

The total action for a single path γ is:

$$S_{\text{entity}}[\gamma, \Psi] = S_{\text{WL}}[\gamma] + S_{\text{spin}}[\Psi] + S_{\text{EM}}[\gamma].$$

3.5 Presence Density

The presence density $\rho(x)$ is defined as:

$$\rho(x) = \int \mathcal{D}[\gamma]\, \mathcal{D}[\Psi]\, e^{\frac{i}{\hbar} S_{\text{entity}}[\gamma,\Psi]}\, \delta^{(4)}(x - \gamma(\tau)).$$

This density encodes how strongly the entity's worldline ensemble occupies spacetime point x.

3.6 Emergent Field

An emergent field $\psi(x)$ can be defined as:

$$\psi(x) = \int d\tau \, \delta^{(4)}(x - \gamma(\tau)),$$

with appropriate spinor structure added via Ψ_a. The standard Dirac Lagrangian:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi,$$

is recovered as the effective field theory of the entity's worldline.

3.7 Curved Spacetime Compatibility

In curved spacetime, the covariant derivative is:

$$D_\mu = \partial_\mu + \Omega_\mu + iqA_\mu,$$

where Ω_μ is the spin connection. SEFI is compatible with this structure, allowing the entity's worldline to propagate in gravitational backgrounds.

4 Interpretation of SEFI

4.1 One Entity, Many Appearances

The electron field is the visible projection of one entity's full path structure. Localized particle detections are local manifestations of the entity's worldline, not separate objects.

4.2 Interference as Path Overlap

Interference patterns arise where different segments of the entity's worldline overlap in spacetime. The interference term in:

$$|\Psi|^2 = |\Psi_{\text{left}}|^2 + |\Psi_{\text{right}}|^2 + 2\Re(\Psi^*_{\text{left}}\Psi_{\text{right}})$$

is interpreted as geometric overlap of worldline segments.

4.3 Measurement as Path Selection

Measurement imposes boundary conditions that restrict the allowed paths. Collapse is:

$$\gamma(\tau) \to \gamma_{\text{selected}}(\tau),$$

a geometric restriction, not annihilation of the entity.

4.4 Electrons and Positrons

Forward-time segments of the worldline appear as electrons; backward-time segments appear as positrons. This is a geometric realization of time-reversed solutions in QFT.

4.5 Spin, Charge, and Fields

Spin arises from Grassmann variables and geometric rotation in internal space. Charge is associated with coupling to A_μ and can be interpreted as a topological invariant of the worldline structure.

5 Superposition in SEFI

5.1 Superposition as Geometry

In standard QM, superposition is a linear combination of states. In SEFI, superposition is the geometric fact that the entity's worldline spans multiple allowed trajectories simultaneously.

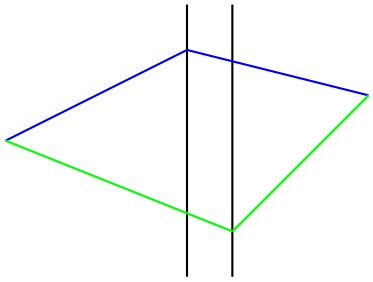
Consider two paths γ_A and γ_B. The total amplitude:

$$\Psi = \Psi_A + \Psi_B$$

corresponds to the entity occupying both paths. This is not a metaphor; it is the literal geometry of the worldline.

5.2 Double-Slit Experiment

In the double-slit experiment, the entity's worldline passes through both slits. The interference pattern on the screen is the projection of the worldline ensemble.



5.3 Collapse as Boundary Condition

When a detector is placed, it imposes a boundary condition that selects one segment of the worldline. The entity remains unified; only the visible manifestation becomes localized.

5.4 Identity and Superposition

Superposition does not violate identity because identity is tied to the entity, not to any single path. The entity is one; its manifestations are many.

6 Geometric Foundations of SEFI

6.1 Entity as Geometric Object

Let M be spacetime and Γ be the entity's full worldline structure:

$$\Gamma : \mathbb{R} \to M.$$

In SEFI, Γ is not a single curve but a geometric set (possibly a manifold, fractal, or fibered structure) whose projection into spacetime yields the electron field.

6.2 Superposition as Geometric Union

Superposition is:

$$\Gamma = \Gamma_A \cup \Gamma_B,$$

a union of worldline segments. Interference is:

$$I(x) = \text{measure}(\Gamma_A \cap \Gamma_B),$$

a geometric intersection.

6.3 Time Reversal as Orientation

Backward-time segments correspond to:

$$\frac{d\Gamma}{d\tau} < 0,$$

an orientation reversal along the worldline. Positrons are time-reversed manifestations of the same entity.

6.4 Spin as Geometric Rotation

Spin can be interpreted as rotation in an internal fiber bundle attached to the worldline. Grassmann variables encode this structure, and geometric phase (Berry phase, holonomy) arises naturally.

6.5 Charge as Topological Invariant

Charge may be associated with topological invariants of Γ, such as winding number or Chern classes, linking SEFI to topological quantum field theory.

6.6 Measurement as Boundary Geometry

Measurement is the intersection:

$$\Gamma \cap \partial M_{\text{detector}},$$

where $\partial M_{\text{detector}}$ is the detector's boundary manifold. Collapse is the restriction of Γ to segments that satisfy this boundary condition.

6.7 Entanglement as Unified Geometry

Entangled systems correspond to a single geometric object:

$$\Gamma_{AB} \subset M,$$

spanning multiple regions. Nonlocal correlations arise because the entity is globally defined, not because of superluminal signaling.

7 Appendix A: Full Mathematical Derivations

7.1 Worldline Formalism

Detailed derivations of the worldline action, path integral measure, and presence density.

7.2 Spinor Construction

Construction of spinor fields from Grassmann variables and worldline supersymmetry.

7.3 Dirac Field Emergence

Derivation of the Dirac equation as the effective field theory of the entity's worldline.

8 Appendix B: Speculative Extensions

8.1 Golden Ratio Weighting

A speculative refinement introduces geometric weighting:

$$w_\varphi[\gamma] = e^{\frac{i}{\hbar}S[\gamma]} \cdot f_\varphi[\gamma],$$

where f_φ encodes self-similar or golden-ratio-related structure in the worldline ensemble.

8.2 Unified Substrate and Consciousness

Philosophical extensions consider the entity as a unified substrate for physical reality and possibly consciousness. These ideas are clearly separated from the core physics.

9 Appendix C: Superposition in SEFI — Deep Interpretation

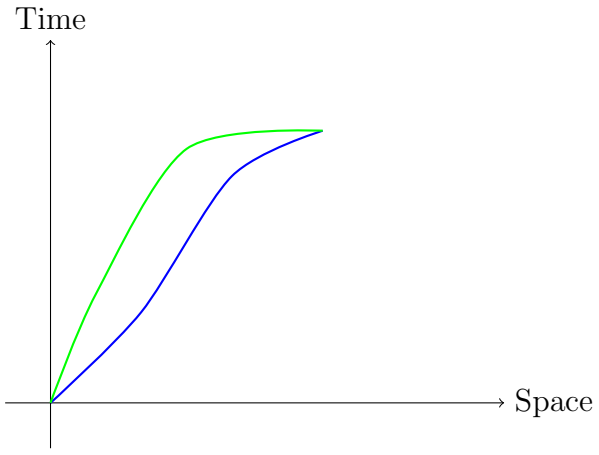
9.1 Overview

Superposition is the distributed manifestation of one entity across multiple trajectories. It is geometric, not merely probabilistic.

9.2 Worldline Multiplicity

$$\rho(x) = \int \mathcal{D}[\gamma]\, e^{\frac{i}{\hbar}S[\gamma]}\, \delta^{(4)}(x - \gamma(\tau)).$$

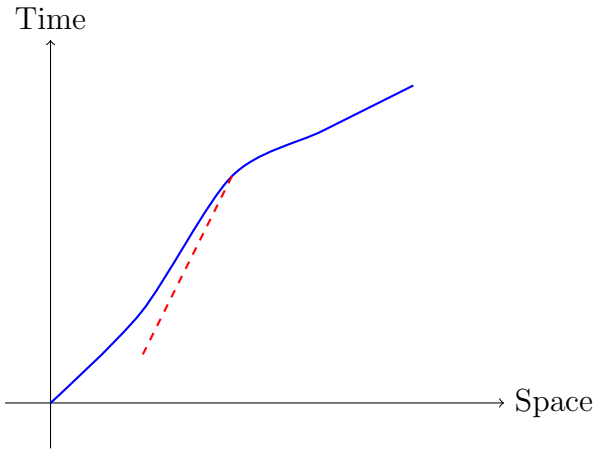
9.3 Geometric Diagram



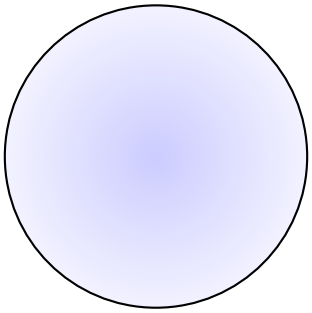
Both paths terminate at the same detection event, illustrating collapse as selection.

10 Diagram Set

10.1 Diagram 1: Worldline



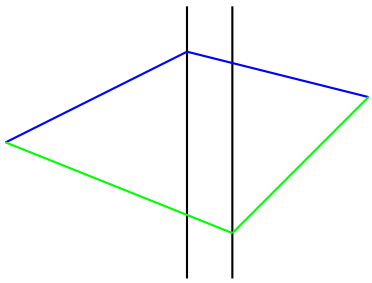
10.2 Diagram 2: Presence Density



10.3 Diagram 3: Time Reversal



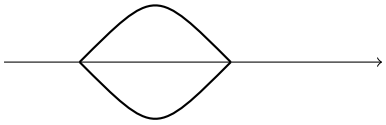
10.4 Diagram 4: Double Slit



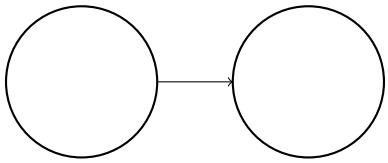
10.5 Diagram 5: Collapse



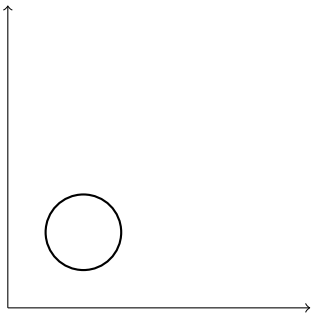
10.6 Diagram 6: Path Integral



10.7 Diagram 7: Identity Preservation



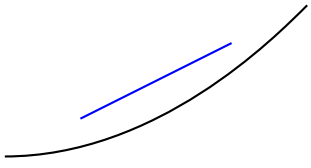
10.8 Diagram 8: Spin Structure



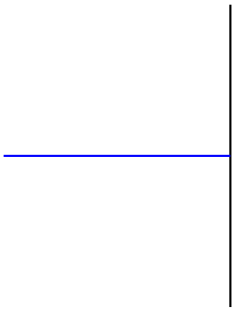
10.9 Diagram 9: Gauge Coupling



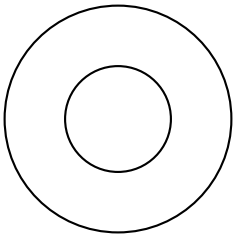
10.10 Diagram 10: Curved Spacetime



10.11 Diagram 11: Measurement Boundary



10.12 Diagram 12: Unified Entity



11 References

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- Schwinger, J. (1951). On gauge invariance and vacuum polarization.
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- Weinberg, S. (1995). The Quantum Theory of Fields.